

11.24

(a.)

Based on the OLS regression output, the variable MIXED is statistically significant at the 5% level, with a p-value of **0.00321**. Furthermore, the joint hypothesis test for STORMY and MIXED yields an F-statistic of **16.14**. Since this F-value is greater than 10, we can conclude that these weather variables are strong instruments for $\ln(\text{PRICE})$.

(b.)

By estimating the demand equation using 2SLS with both STORMY and MIXED as instrumental variables, the new price elasticity of demand ($\ln(\text{PRICE})$ coefficient) is -0.93014. Compared to the estimate of -1.1194 in Table 11.5 (which only used STORMY as an IV), the addition of MIXED has changed the magnitude of the elasticity and decreased its standard error, potentially improving the precision of our estimate.

(c.)

The Sargan test for overidentifying restrictions yields a p-value of 0.380. Because this p-value is greater than 0.05, we fail to reject the null hypothesis. Therefore, we conclude that the instruments (STORMY and MIXED) are valid and correctly excluded from the demand equation.

(d.)

The joint F-test for the daily indicator variables (MON, TUE, WED, THU) in the reduced-form equation results in an F-statistic of 0.6279 and a p-value of 0.6437. Since the p-value is > 0.05 and the F-value is far below 10, these variables are jointly insignificant. Consequently, we conclude that we **cannot** reliably estimate the supply equation by 2SLS, as the demand shifters act as weak instruments, meaning the supply curve is not well-identified in practice.

11.28

a.

$$D: Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 DI_i + e_{di}$$

$$S: Q_i = \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_{si}$$

Inverse Demand Equation: $P_i = -\frac{\alpha_1}{\alpha_2} + \frac{1}{\alpha_2} Q_i - \frac{\alpha_3}{\alpha_2} PS_i - \frac{\alpha_4}{\alpha_2} DI_i - \frac{1}{\alpha_2} e_{di}$

where $(\frac{1}{\alpha_2})$ is expected to be negative.
 $(-\alpha_3/\alpha_2)$ is expected to be positive.
 $(-\alpha_4/\alpha_2)$ is expected to be positive.

Inverse Supply Equation: $P_i = -\frac{\beta_1}{\beta_2} + \frac{1}{\beta_2} Q_i - \frac{\beta_3}{\beta_2} PF_i - \frac{1}{\beta_2} e_{si}$

where $(\frac{1}{\beta_2})$ is expected to be positive.
 $(-\beta_3/\beta_2)$ is expected to be positive.

b.

```
ivreg(formula = p ~ q + ps + di | ps + di + pf, data = truffle
s)

Residuals:
    Min       1Q   Median       3Q      Max
-39.661  -6.781   2.410   8.320  20.251

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -11.428     13.592  -0.841  0.40810
q             -2.671      1.175  -2.273  0.03154 *
ps              3.461      1.116   3.103  0.00458 **
di             13.390      2.747   4.875  4.68e-05 ***
```

The estimated coefficient for Q is Negative (expected negative), the coefficient for PS is Positive (expected positive), and the coefficient for DI is Positive (expected positive). These align perfectly with the anticipated signs derived in part (a). Based on the regression output, the p-values for Q, PS, and DI are 0.03154, 0.00458, and 0.0000468, respectively. Since these are less than 0.05, the estimated coefficients are statistically significant at the 5% level.

```
ivreg(formula = p ~ q + pf | pf + ps + di, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-9.7983 -2.3440 -0.6281  2.4350 11.1600

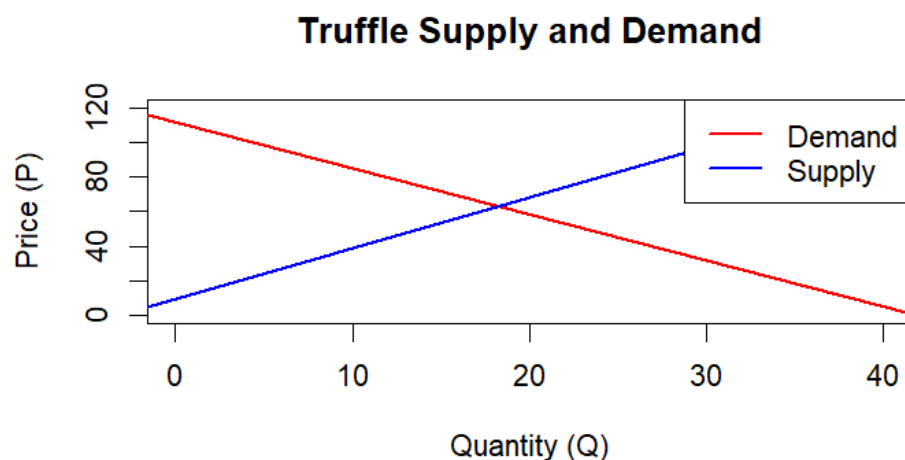
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -58.7982     5.8592  -10.04 1.32e-10 ***
q             2.9367     0.2158   13.61 1.32e-13 ***
pf            2.9585     0.1560   18.97 < 2e-16 ***
```

The estimated coefficient for Q is Positive(expected positive), and the coefficient for PF is Positive (expected positive). These match the theoretical expectations from part (a). The p-values for Q and PF are less than 0.05, so the coefficients are statistically significantly different from zero.

c.

Price Elasticity of Demand at the means is: -1.2723"

d.



e.

```
> cat("Reduced-form 預測均衡數量 Q* =", Q_reduced, "\n")
Reduced-form 預測均衡數量 Q* = 18.2601
> cat("Reduced-form 預測均衡價格 P* =", P_reduced, "\n\n")
Reduced-form 預測均衡價格 P* = 62.81425

> cat("Structural (Part d) 預測均衡數量 Q* =", Q_structural,
"\n")
Structural (Part d) 預測均衡數量 Q* = 18.25021
> cat("Structural (Part d) 預測均衡價格 P* =", P_structural,
"\n")
Structural (Part d) 預測均衡價格 P* = 62.84257
```

The equilibrium values obtained from part (d) (solving the estimated 2SLS structural equations) and the values obtained from the OLS reduced-form equations in Table 11.2 are virtually identical, subject only to minor rounding errors in the reported coefficients. This agreement occurs because the reduced-form equations are derived algebraically by solving the simultaneous structural equations for the endogenous variables (P and Q) as functions of solely the exogenous variables. Therefore, evaluating both methods at the same exogenous values (PS, DI, PF) must yield the same equilibrium point.

f.

```
lm(formula = p ~ q + ps + di, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-25.0753  -2.7742  -0.4097   4.7079  17.4979

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.6195     9.0872  -1.499   0.1460
q             0.1512     0.4988   0.303   0.7642
ps            1.3607     0.5940   2.291   0.0303 *
di            12.3582     1.8254   6.770 3.48e-07 ***
```

```
lm(formula = p ~ q + pf, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-8.4721 -3.3287   0.1861   2.0785  10.7513

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -52.8763     5.0238  -10.53 4.68e-11 ***
q             2.6613     0.1712   15.54 5.42e-15 ***
pf            2.9217     0.1482   19.71 < 2e-16 ***
```

Attempting to estimate the structural equations using OLS produces severely biased results, including an incorrect positive slope for the demand curve (the sign is not correct). This occurs because Price and Quantity are jointly determined, causing **simultaneous equations bias (endogeneity)**. This clearly proves that **2SLS is absolutely necessary** to accurately estimate market supply and demand models.